IMT SCUOLA ALTI STUDI LUCCA

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Near-Codewords Aware Bit Flipping Decoding of QC-MDPC Codes

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Post-quantum cryptography

Problem

Cryptographic schemes that cannot be attacked efficiently by *classical* nor *quantum algorithms*.

Main sources of hard problems for post-quantum cryptography:

- Polynomial systems
- Lattices
- Isogenies between supersingular elliptic curves
- **Coding Theory**

Code-Based primitives

Schemes in the 1° NIST PQC competition:

- *1° round:* 69 candidates (19 code-based)
- *2° round:* 26 candidates (7 code-based)
- *3° round:* 15 candidates (3 code-based)
- *4° round:* 3 code-based `BIKE` [1] - `ClassicMcEliece` [2] - `HQC` [3]

NIST Outcome

- `HQC` ⇒ ✓ Selected
- `Classic McEliece` ⇒ ✗ Not selected
- `BIKE` ⇒ ✗ Not selected

Code-Based primitives

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> *"NIST reviewed several Decryption Failure Rate (DFR) analyses of BIKE, ⋯ . Despite these promising results, NIST found the security analysis of HQC to be more mature and stable than that of BIKE. As such, NIST has not selected BIKE for standardization."*
>
> NIST IR 8545, 2025

What is the DFR and why is so important?

The issue of estimating DFR

Decoding failure ⇒ Decryption failure ⇒ Leaks secret key info

Adaptively Chosen Ciphertext Attack (CCA2): adversaries can query a decryption oracle.

IND-CCA2 security requires DFR$< 2^{-\lambda}$, with $\lambda \geq 128$ [4]

Decoding Failure Rate

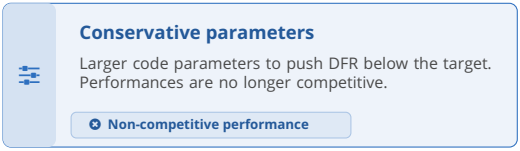
In BIKE:

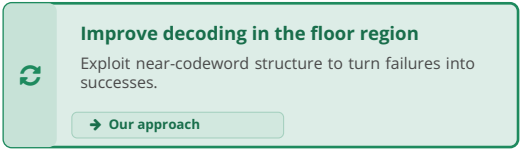
- **H**: random QC-MDPC parity check matrix (defined later)
- e: random error vector with weight t

For Dec a decoder, the DFR is the probability that a decoding failure happens (i.e., that Dec($\mathbf{H}$, $\mathbf{He}^\top$) $\neq$ **e**), averaged over all parity-check matrices and all error vectors.

Our Contribution

At low DFR main events of decoding failures are due to near-codewords [5].
What can we do to achieve the target security level?





Main Contributions

- Exploit the knowledge of near-codewords in the decoding phase
- The way we use this knowledge works with any BF decoder
- Best results are achieved with BF-Max

Notation

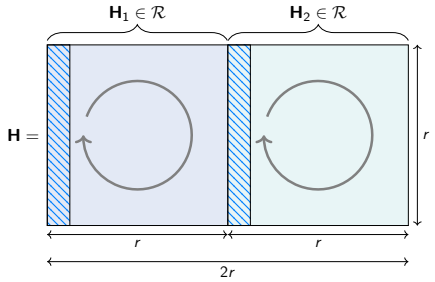
Double circulant QC-MDPC codes

Let $\mathbf{H} = (\mathbf{H}_1,\ \mathbf{H}_2) \in \mathcal{R}^2$, $\mathbf{H}_1$ and $\mathbf{H}_2$ with weight $v = O(\sqrt{r})$ and

$$\mathscr{C} = \left\{ \mathbf{c} \in \mathbb{F}_2^{2r} \mid \mathbf{H}\mathbf{c}^\top = \mathbf{0} \right\}$$

We say that $\mathscr{C}$ is a double circulant QC-MDPC code.

- r : circulant side
- v : fixed circulant weight
- t : fixed error weight
- BF: Bit Flipping decoder
- $\mathcal{R} \subseteq \mathbb{F}_2^{r \times r}$: ring of $r \times r$ circulant matricies



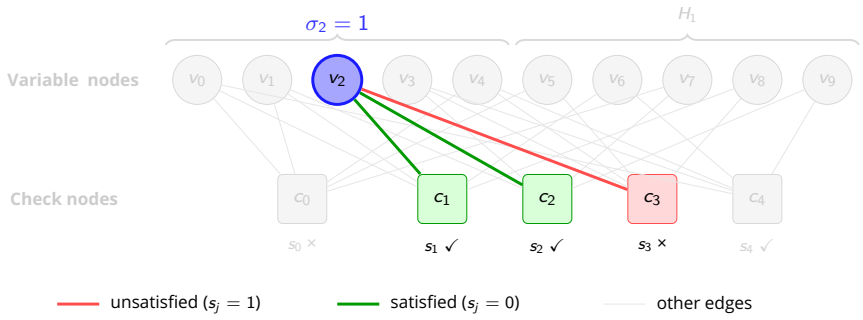
Counters

Definition

For $i \in \{1, \cdots, n\}$, the i-th counter is the number of unsatisfied parity-check equations in which the i-th bit participates. Note that

$$\sigma_i \in \{0,\ 1,\ \cdots,\ v\}, \quad \forall i$$

Rule of thumbs: if $e_i = 1$, then σ_i "should" be high & if $e_i = 0$, then σ_i "should" be low



BF Decoding

- Estimate the error vector in an iterative fashion
- Good trade-off between complexity, error correction capability and simplicity

BF($\mathbf{H}$, $\mathbf{s}$, BFsetting)

- $\widehat{\mathbf{e}} \leftarrow \mathbf{0}$ \\Initially null error estimate
- $\mathtt{Iter} \leftarrow 0$ \\Number of performed iterations
- While $(\mathbf{s} \neq \mathbf{0}) \vee (\mathtt{Iter} < \mathtt{IterMax})$:
  - Compute $(\sigma_1, \cdots, \sigma_n)$ \\Counters for all bits
  - Determine J : set of indices with large counters \\Bits to flip
  - For $i \in J$:
    - $\widehat{e}_i \leftarrow \widehat{e}_i + 1$ \\Update error estimate
    - $\mathbf{s} \leftarrow \mathbf{s} + \mathbf{h}_i$ \\Update syndrome
  - $\mathtt{Iter} \leftarrow \mathtt{Iter} + 1$ \\Update number of iterations
- If $\mathbf{s} \neq \mathbf{0}$ return $\perp$, else return $\widehat{\mathbf{e}}$

BF variants: how many bits are flipped, how they are selected, ...

Near-codewords

Well known example of Trapping sets for QC- MDPC codes.

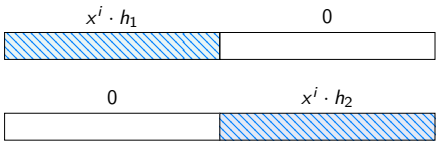
Special error vectors $\Rightarrow$ *unexpected behavior* $\Rightarrow$ *Decoding failure*

Definition

Let $\mathbf{H} = (h_1, h_2)$ *be the parity-check matrix of a double circulant code;*
then the set of near-codewords is

$$\mathcal{E} = \left\{ (x^i \cdot h_1, 0) \;\middle|\; i = 0, \cdots, r-1 \right\} \cup \left\{ (0, x^i \cdot h_2) \;\middle|\; i = 0, \cdots, r-1 \right\}.$$

$$\text{if} \quad \mathbf{e} = (x^i \cdot h_1, 0) \quad \text{then} \quad \mathbf{s} = \mathbf{e}\mathbf{H}^\top = x^i \cdot h_1 \cdot h_1 + 0 \cdot h_2 = x^i \cdot h_1^2 \implies hw(\mathbf{s}) = v$$



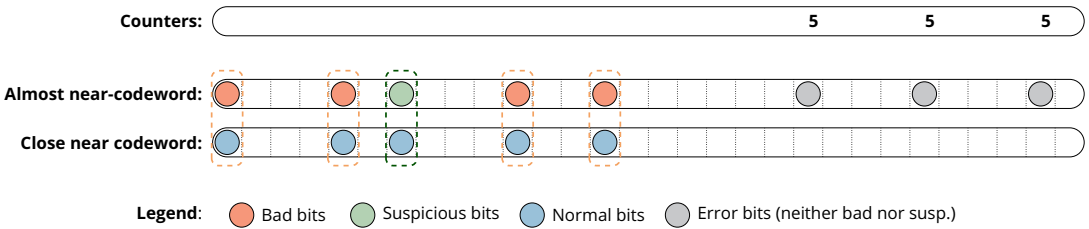
Convergence to near-codewords

Definition

For a matrix $\mathbf{H}$ for which $\mathcal{E}$ is the set of near-codewords, let $\mathcal{E}_{t,u} \subseteq \mathbb{F}_2^n$ be the set of t-weight vectors such that, for each $\mathbf{e} \in \mathcal{E}_{t,u}$, it holds that

$$\max_{\mathbf{m} \in \mathcal{E}} \{|\mathrm{Supp}(\mathbf{e}) \cap \mathrm{Supp}(\mathbf{m})|\} = u$$

- We refer to elements of $\mathcal{E}_{t,u}$ as (t, u)-*almost near-codewords*.
- We refer to $\mathbf{m}$ as the *close near-codeword* of $\mathbf{e}$.



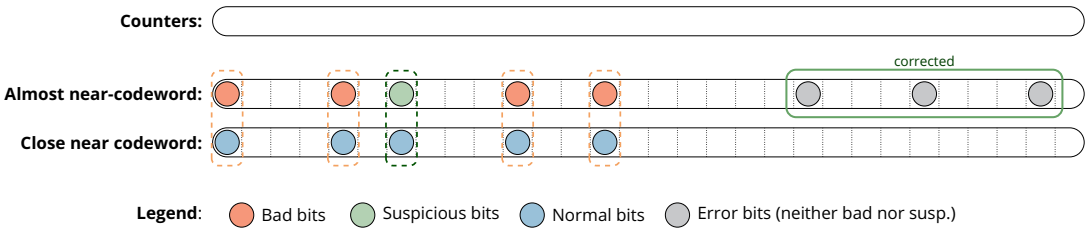
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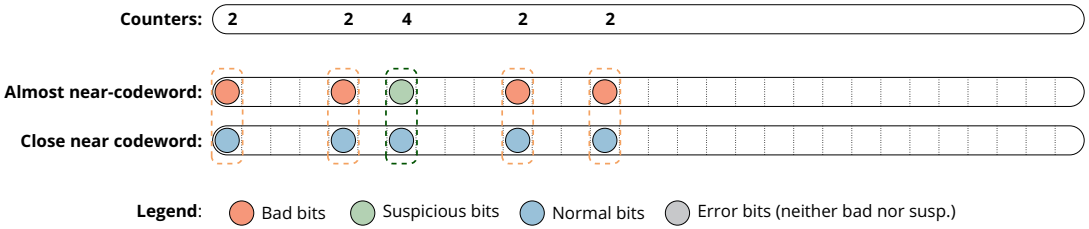
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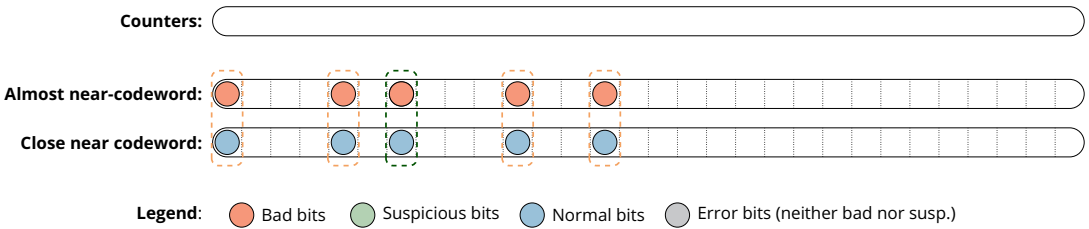
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Near-codewords aware decoding

Can we exploit the knowledge of this errors? Yes!!

The idea in a nutshell

Store near-codewords, recognize when BF gets stuck and escape!

- Precomputed and store

$$\texttt{NC_Syndromes} = \left\{ \left(\mathbf{m}_i \mathbf{H}^\top, i \right) \mid i = 0, \cdots, 2r-1 \right\},$$

- Compare at each iteration the current syndrome with the precompute ones
- If yes we fall in a near codewords.
- But now we can recover the bad situation.

Near-codewords aware decoding

The changes that need to be made to the decoder are minimal

BF($\mathbf{H}$, $\mathbf{s}$, BFsetting)

- $\widehat{\mathbf{e}} \leftarrow \mathbf{0}$ \\Initially null error estimate
- Iter $\leftarrow 0$ \\Number of performed iterations
- While $(\mathbf{s} \neq \mathbf{0}) \vee (\texttt{Iter} < \texttt{IterMax})$:
  - Compute $(\sigma_1, \cdots, \sigma_n)$ \\Counters for all bits
  - Determine J : set of indices with large counters \\Bits to flip
  - For $i \in J$:
    - $\widehat{e}_i \leftarrow \widehat{e}_i + 1$ \\Update error estimate
    - $\mathbf{s} \leftarrow \mathbf{s} + \mathbf{h}_i$ \\Update syndrome
  - If(hw(s) = v) then \\Check if we are falling in a trapping sets
    - NC_Lookup \\Lookup inside the precomputed table
    - $\widehat{e} \leftarrow \widehat{e} + m$ \\Use the knowledge to update the error
  - Iter $\leftarrow$ Iter $+ 1$ \\Update number of iterations
- If $\mathbf{s} \neq \mathbf{0}$ return $\perp$, else return $\widehat{\mathbf{e}}$

Results

Asses the improvement of our modification through simulations.

- Simulate only decoding of (t, u)-almost near-codewords, for several values of u.
- For each decoder up to 10^6 simulations for each values of u.
- Multi thread decoder in C, source code available at `github.com/secomms/BF_near_codewords`

Simulations

BIKE parameters:

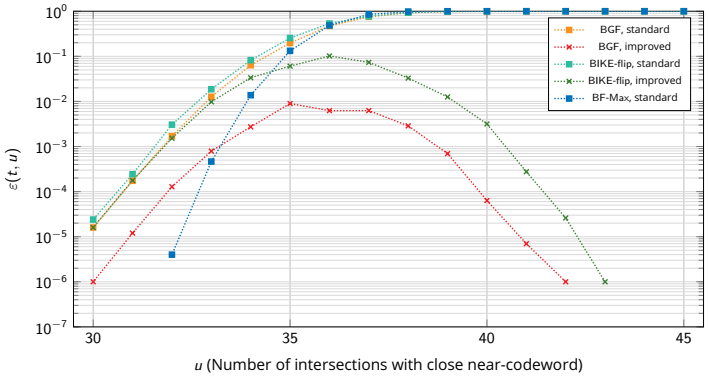
- $t = 134$
- $v = 71$
- $p = 12323$
- $n = 24646$

Main cryptographic decoders:

		Type	`IterMax`	Complexity
BGF	[6]	Threshold	5	$2^{25.70}$
BIKE-flip	[1]	Adaptive Threshold	7	$2^{26.19}$
BF-Max	[7]	Max counter	268 ($2t$)	$2^{25.69}$

Results

- The probability of failure has decreased significantly
- Especially true for large values of u
- BF-Max seems to be the decoder with the lowest failure probability.



Conclusions

- Our modification works with any BF decoder
- Very mild time complexity $O\left(\sqrt{r} \cdot \log_2^2(r)\right)$ compared to the decoder complexity $O\left(r^{1.5} \cdot \log_2(r)\right)$
- BF-Max has the lowest observable DFR among state-of-the-art decoders
- BF-Max exhibits the lowest complexity

Future works

Approximation of the DFR in the extended version

Decoder	DFR	Improved DFR
BGF	$\approx 2^{-165}$	$\approx 2^{-169}$
BIKE-flip	$\approx 2^{-165}$	$\approx 2^{-165}$
BF-Max	$\approx 2^{-183}$	$\approx 2^{-231}$

Table: DFR approximations for BIKE Cat. 1 parameters.

Acknowledgements

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Thank You!

Questions?

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